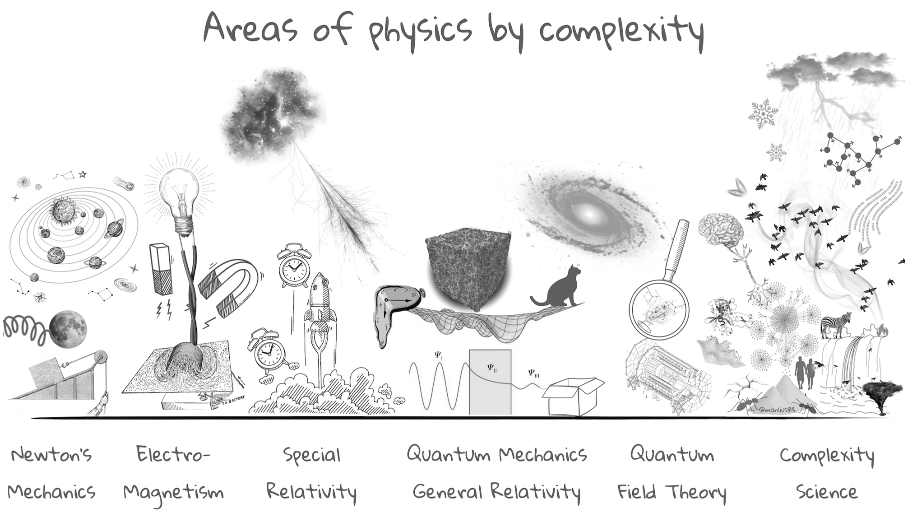
Final Report

Physics of Complex Networks: Structure and Dynamics

UNIVERSITÀ DEGLI STUDI DI PADOVA



Projects Report: #15, #44

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1 | Task 15 - SOC and Cascade Optimization in Coupled Sandpiles

1.1 | Theoretical Introduction

The primary objective of this investigation is to identify the optimal interconnectivity threshold that minimizes global risk in coupled infrastructures, balancing the trade-offs that govern cascades of load shedding.[3] While establishing links between independent networks allows a critical system to initially divert excess load to a neighboring module acting as a reservoir, excessive coupling paradoxically accelerates systemic vulnerability by opening open pathways for destructive, cross-module cascade propagation.[3] To quantitatively evaluate these dynamics, the **Bak-Tang-Wiesenfeld (BTW)** sandpile formulation is simulated on a multitype network framework composed of two interacting modules, A and B, generated via a regular configuration model where each node possesses an identical internal degree $z_a = z_b = 3$ and a total system size of $N = 2000$ vertices per module.[3]

The driving mechanism introduces units of load stochastically and uniformly across the nodes.[3] Every vertex i is characterized by an innate capacity threshold equal to its total degree k_i.[3] Whenever the accumulated load h_i strictly exceeds this capacity, the node becomes unstable and undergoes a shedding event, transferring exactly one unit of load to each of its k_i neighbors.[3] The balancing transport equation governing the redistribution of load from a triggered node i to its adjacent neighborhood $\mathcal{N}(i)$ is formulated as:[2]

$$\begin{cases} h_i \longrightarrow h_i - k_i \\ h_j \longrightarrow h_j + 1 \quad \forall\, j \in \mathcal{N}(i) \end{cases} \tag{1.1}$$

To establish a stationary critical state and avoid indefinite mass accumulation on finite graphs, a uniform dissipation rate f is integrated into the propagation rule.[3] During each shedding event, individual grains are independently deleted from the system with a small probability f as they traverse the links, effectively controlling the characteristic exponential cutoff scale of the largest global cascades.[3]

1.2 | Network Topology and Simulation Setup

The modular topology is assembled by implementing a Bernoulli-distributed coupling mechanism to establish sparse interdependencies between the regular baselines.[3] Under this multitype configuration, each node in module A receives exactly one external edge stub with a coupling probability p, and zero stubs with probability $1-p$.[3] These external stubs are subsequently matched and wired uniformly at random with a corresponding sequence of stubs generated independently in module B.[3] From a structural standpoint, the creation of an inter-module bridge increments both the local degree k_i and the capacity threshold of the endpoint vertex by one unit.[3]

The numerical analysis explores the transition from microscopic stochastic events to macroscopic collective vulnerabilities by testing how the network interconnection alters the total cascade distribution profiles.[3] The statistics of the cascading events are evaluated using the complementary cumulative distribution function (CCDF), which filters out statistical noise in the distribution tails. Figure 1.1 outlines the microscopic properties of the cascade ensembles recorded over 5×10^5 driving steps after discarding a transient loading phase of 5×10^4 steps.

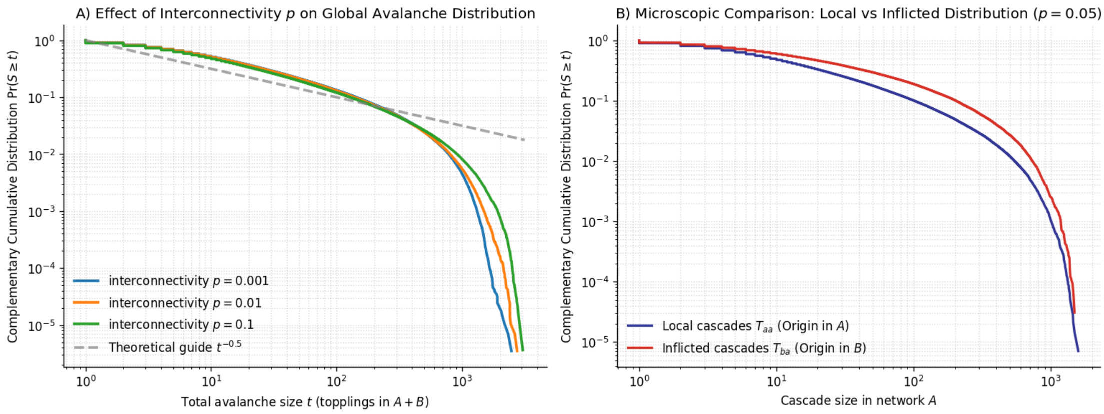

Figure 1.1: (Left) CCDF log-log distribution of the total cascade size t $(A + B)$ for three coupling regimes ($p = 0.001, 0.01, 0.1$), compared against the mean-field scaling guide $t^{-0.5}$. (Right) Microscopic comparison between the localized cascade sizes T_{aa} (origin in A) and the exogenous inflicted cascades T_{ba} (origin in B) at a fixed coupling level $p = 0.05$.

For intermediate scaling windows, the empirical data match the classic mean-field power-law exponent ($\tau = 3/2$, implying a CCDF slope of -0.5) typical of sparse, locally tree-like random regular topologies where the branching process approximation becomes exact.[3, 6] As the interconnectivity p increases from 0.001 to 0.1, the total available capacity scales up, shifting the exponential cutoff toward larger cascade sizes t and enabling system-wide global propagation.[3] Isolating the directional flow of risk at $p = 0.05$, the distribution tail of the inflicted cascades (T_{ba}) decays slower than that of the purely local events (T_{aa}).[3] This diagnostics validates the structural role of inter-module bridges, which act as efficient conduits for cross-module load transfer, amplifying external load shocks.[3]

1.3 | Macroscopic Phase Space and Stability

Integrating the statistical event counts, the systemic risk is projected onto a macroscopic phase space to trace the emergence of an optimal network topology.[3] A large cascade is defined via a threshold cutoff $C = N/2 = 1000$ nodes.[3]

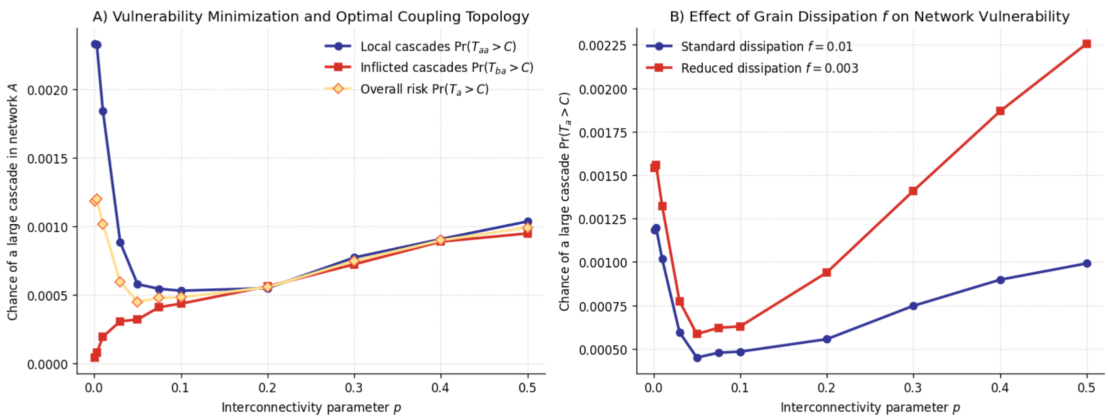

Figure 1.2: (Left) Linear phase space of the large cascade probabilities ($\Pr(T > 1000)$) as a function of the interconnectivity p, highlighting the decomposition into local, inflicted, and overall risk. (Right) Sensitivity analysis illustrating the overall cascading risk profile under standard ($f = 0.01$) and reduced ($f = 0.003$) grain dissipation.

The linear phase space outlined in Figure 1.2 maps a non-monotonic vulnerability profile.[3] At small p, increasing the number of bridges suppresses local cascades ($\Pr(T_{aa} > C)$ drops) because the neighboring network acts as a dissipating reservoir that absorbs excess load.[3] Conversely, at high p, strong coupling increases capacity and enables inflicted cascades ($\Pr(T_{ba} > C)$ rises), letting fluctuations return with catastrophic effects.[3] To extract the continuous minimum independent of the discrete grid points, a local second-degree polynomial windowing is applied within the sella region ($p \in [0.01, 0.15]$), yielding an optimal coupling point at $p^* \approx 0.0717$, in agreement with the analytical predictions (0.075 ± 0.01).[3]

A sensitivity analysis checks the structural vulnerability of the phase space against changes in the conservation rules. Under a reduced dissipation rate ($f = 0.003$), the system retains a larger mass fraction per shedding event.[3] The entire vulnerability profile undergoes a vertical translation, amplifying the frequency of large events and accelerating the catastrophic risk ascent at high p, while preserving the geometric localization of the optimal coupling minimum.[3]

2 | Task 44 - Social Connectedness Index II from Facebook

2.1 | Theoretical Introduction

This analysis investigates how geographical distance, administrative boundaries, and socio-economic factors structurally shape online communities through the lens of spatial network theory. The empirical characterization of these sub-national social networks is conducted utilizing the **Meta Social Connectedness Index (SCI) II** database, which measures the relative friendship probability between administrative locations i and j defined as:[1]

$$SCI_{i,j} = \frac{\text{Friendships}_{i,j}}{\text{Pop}_i \times \text{Pop}_j} \tag{2.1}$$

where $\text{Friendships}_{i,j}$ represents the total number of Facebook links between the two regions, and $\text{Pop}_i, \text{Pop}_j$ reflect the respective cohorts of active users.[1] By filtering out cross-border connections, this work inspects administrative fragmentation, spatial mixing via a distance-decay framework, and national structural regimes—ranging from centralized hubs to polycentric topologies—based on network size and strength heterogeneity metrics.

2.2 | Dataset and Network Construction

The pipeline retains exclusively within-country edges, filtering the top 100 nations by sub-national region count.[8] The United States (US) are excluded to avoid macro-scaling distortions. The target countries are split into two layers based on data availability: European territories mapped to the **NUTS3** framework and non-European territories mapped to **GADM Level 1**.

Shapefiles are downloaded from GADM for the non-European layer,[5] and the NUTS 2024 GeoJSON is gathered from Eurostat.[4] Spatial coordinates are computed as the representative points of the regional geometries. Symmetric duplicates are averaged into undirected edges, and self-loops are omitted. The final parsed network integrates both layers, yielding $N = 3,147$ nodes and $E = 136,844$ edges.

The distribution of regions per country shows a severe fragmentation asymmetry captured by a lognormal PDF fit: most nations have fewer than 30 subdivisions, while a few exceed 100. Germany (DEU) is the main outlier with 399 nodes due to its granular NUTS3 framework.

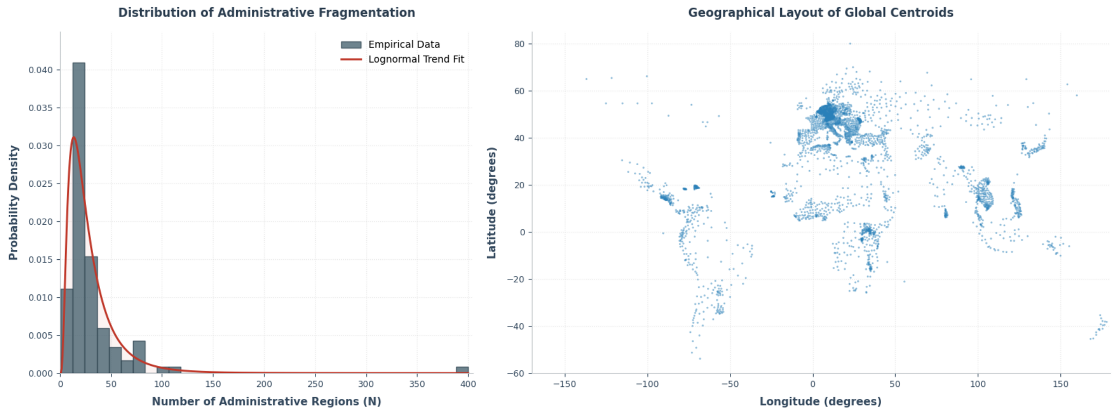

Figure 2.1: (Left) Distribution of administrative fragmentation (N) fitted with a Lognormal PDF trend. (Right) Geographical layout of global regional centroids computed from shapefile geometries.

2.3 | Network Data Analytics

A quadratic scaling law links the number of edges to the network size ($E \sim N^{2.03}$, $R^2 = 1.00$), meaning the networks are topologically complete cliques with an unweighted density $\rho = 1$. The real degrees of freedom are stored within the link intensities. The probability density of $\log_{10}(SCI)$ is heavily right-skewed and spans six orders of magnitude, separating a dense backbone of weak interactions from a narrow tail of intense regional flows.

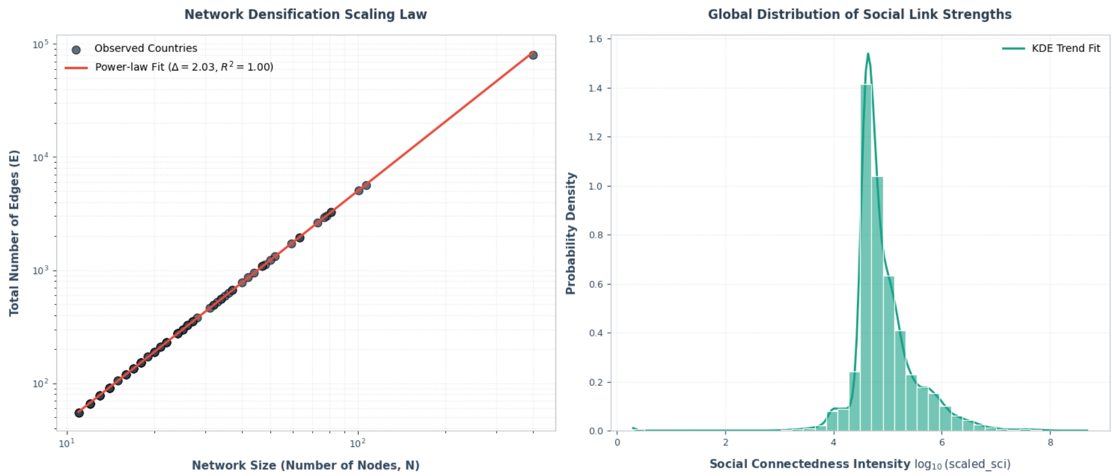

Figure 2.2: (Left) Network densification scaling law mapping N vs E in log-log scale. (Right) Global probability density distribution of the social link weights.

Integrating geospatial coordinates, we fit the link intensities to a power-law distance-decay Gravity Law using Haversine distance: $SCI_{ij} \sim d_{ij}^{-\alpha}$. The linear fit yields a global exponent $\alpha = 0.78$, demonstrating that virtual connections decrease almost inversely with distance. The highest density of connections is packed within the 100–600 km range, while long-range tails extend beyond $10,000$ km.

Network organization is summarized via the Gini coefficient ($G \in [0, 1]$) on node strengths $\{s_i\}$.[7] Mapping countries into a phase space defined by size and heterogeneity relative to their global medians yields four structural regimes: *Small and Concentrated*, *Large and Concentrated*, *Small and Homogeneous*, and *Large and Homogeneous*. The upper right quadrant represents centralized topologies with a dominant hub, like Colombia (COL, $G > 0.30$). The bottom right represents fragmented yet uniform networks, like Japan (JPN, $G \approx 0.07$). The bottom left quadrant accommodates small, homogeneous networks like Jamaica (JAM).

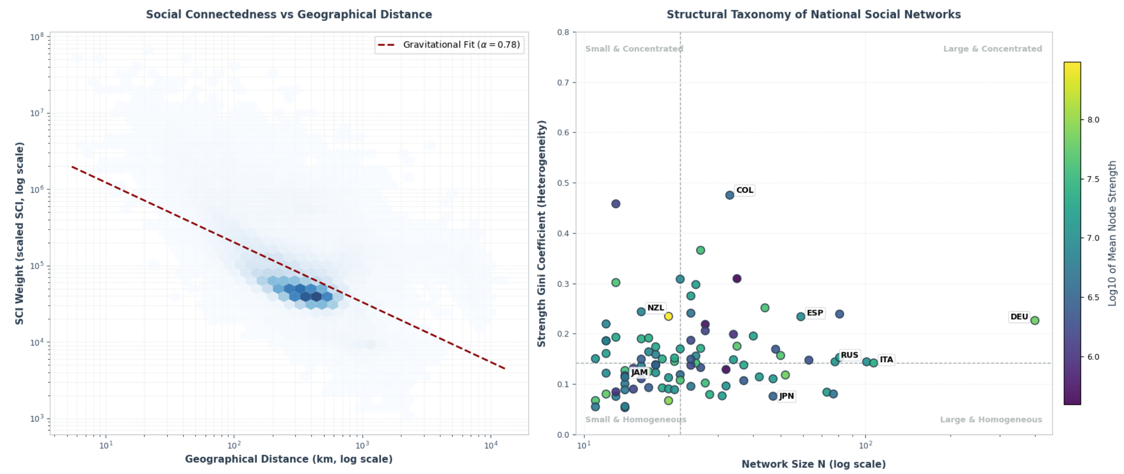

Figure 2.3: (Left) Empirical scaling of SCI weight as a function of Haversine geographical distance fitted via log-log linear regression. (Right) Structural taxonomy phase space of national social networks based on size N and strength Gini coefficient.

Italy occupies a borderline position exactly on the global Gini median line ($G \approx 0.14$). The economic dominance of major centers (Milan, Rome) acts as a centripetal force pulling the Gini upward. However, this concentration is mitigated by a decentralized, polycentric regional structure with strong secondary nodes and interconnected clusters. This polycentric equilibrium anchors the Italian macro-social network in a stable transition state between homogeneity and centralization.

3 | Bibliography

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